

# K-FoldGridDistilBert\_Multiclassification

April 15, 2026

```
[1]: # Install required packages
!pip install -q transformers scikit-learn matplotlib seaborn
```

[notice] A new release of pip is  
available: 25.3 -> 26.0.1

[notice] To update, run:  
pip install --upgrade pip

```
[2]: # =====
# DistilBERT - Leakage-Free Pipeline (Aligned with BiGRU/CNN)
# 1. Split: Train (85%) + Test (15%) + test frozen
# 2. Grid search via 3-Fold CV on Train (85%)
# 3. Final retrain on full oversampled Train (85%)
# 4. Evaluate once on Test (15%)
# =====

import os
import random
import re
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, StratifiedKFold
from sklearn.metrics import classification_report, confusion_matrix
from imblearn.over_sampling import RandomOverSampler # Added for consistency

import torch
from torch.utils.data import Dataset, DataLoader
from torch.optim import AdamW
from transformers import (
    DistilBertTokenizerFast,
    DistilBertForSequenceClassification,
    get_linear_schedule_with_warmup
)
```

```

# --- Reproducibility ---
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
torch.manual_seed(SEED)
if torch.cuda.is_available():
    torch.cuda.manual_seed_all(SEED)
device = torch.device("mps") if torch.backends.mps.is_available() else torch.
    ↪device("cpu")

# --- Load Data ---
df = pd.read_csv('annotated_dataset_for_multi_classifications.csv',
    ↪encoding='utf-8-sig')

# --- Class Mapping (unchanged) ---
y_dict = {
    'functionality issue': 0,
    'user experience issue': 1,
    'performance issue': 2,
    'stability issue': 3,
    'bug': 4,
    'device issue': 5,
    'customer support issue': 6,
    'privacy issue': 7,
    'user interface issue': 8,
    'compatibility issue': 9,
    'installation issue': 10,
    'security issue': 11,
    'network issue': 12
}
class_labels = [k for k in y_dict.keys()]
n_classes = len(class_labels)

df['category'] = df['category'].str.strip().str.lower()
df = df[df['category'].isin(y_dict)]
df['label_id'] = df['category'].map(y_dict).astype(int)

print("Classes (fixed order):")
for i, name in enumerate(class_labels):
    print(f"{i} → {name}")

# --- REVISED SPLIT: 85% Train / 15% Test (NO VAL SET) ---
texts = df['Base_Reviews'].values
y_int = df['label_id'].values

X_train_raw, X_test_raw, y_train_raw, y_test = train_test_split(
    texts, y_int, test_size=0.15, random_state=SEED, stratify=y_int

```

```

)

print(f"\n Final Splits: Train (85%): {len(X_train_raw)}, Test (15%): {len(X_test_raw)}")

# --- Tokenizer (unchanged - static, no leakage risk) ---
tokenizer = DistilBertTokenizerFast.from_pretrained('distilbert-base-uncased')

def tokenize_texts(texts, max_len):
    return tokenizer(
        [str(t) for t in texts],
        padding='max_length',
        truncation=True,
        max_length=max_len,
        return_tensors='pt',
        return_attention_mask=True
    )

# --- Dataset wrapper (unchanged) ---
class TextDS(Dataset):
    def __init__(self, toks, labels):
        self.ids = toks['input_ids']
        self.msk = toks['attention_mask']
        self.y = torch.tensor(labels, dtype=torch.long)

    def __len__(self):
        return len(self.y)

    def __getitem__(self, i):
        return {
            'input_ids': self.ids[i],
            'attention_mask': self.msk[i],
            'labels': self.y[i]
        }

# --- Train/Eval utilities (unchanged) ---
def train_one_run(train_loader, val_loader, epochs, lr, weight_decay=0.01):
    model = DistilBertForSequenceClassification.from_pretrained(
        'distilbert-base-uncased', num_labels=n_classes
    ).to(device)

    optimizer = AdamW(model.parameters(), lr=lr, weight_decay=weight_decay)
    total_steps = len(train_loader) * epochs
    scheduler = get_linear_schedule_with_warmup(
        optimizer, num_warmup_steps=0, num_training_steps=total_steps
    )

```

```

best_val_acc = -1.0
best_state = None
history = {'train_loss': [], 'val_loss': [], 'train_acc': [], 'val_acc': []}

for ep in range(epochs):
    # Train
    model.train()
    tr_loss, tr_correct, tr_seen = 0.0, 0, 0
    for batch in train_loader:
        optimizer.zero_grad()
        inputs = batch['input_ids'].to(device)
        masks = batch['attention_mask'].to(device)
        labels = batch['labels'].to(device)
        outputs = model(inputs, attention_mask=masks, labels=labels)
        loss = outputs.loss
        loss.backward()
        optimizer.step()
        scheduler.step()

        tr_loss += loss.item()
        preds = torch.argmax(outputs.logits, dim=1)
        tr_correct += (preds == labels).sum().item()
        tr_seen += labels.size(0)

    avg_tr_loss = tr_loss / len(train_loader)
    avg_tr_acc = tr_correct / max(1, tr_seen)
    history['train_loss'].append(avg_tr_loss)
    history['train_acc'].append(avg_tr_acc)

    # Validate
    model.eval()
    val_loss, val_correct, val_seen = 0.0, 0, 0
    with torch.no_grad():
        for batch in val_loader:
            inputs = batch['input_ids'].to(device)
            masks = batch['attention_mask'].to(device)
            labels = batch['labels'].to(device)
            outputs = model(inputs, attention_mask=masks, labels=labels)
            val_loss += outputs.loss.item()
            preds = torch.argmax(outputs.logits, dim=1)
            val_correct += (preds == labels).sum().item()
            val_seen += labels.size(0)

    avg_val_loss = val_loss / len(val_loader)
    avg_val_acc = val_correct / max(1, val_seen)
    history['val_loss'].append(avg_val_loss)
    history['val_acc'].append(avg_val_acc)

```

```

        if avg_val_acc > best_val_acc + 1e-6:
            best_val_acc = avg_val_acc
            best_state = {k: v.detach().cpu().clone() for k, v in model.
↪state_dict().items()}

    if best_state is not None:
        model.load_state_dict({k: v.to(device) for k, v in best_state.items()})
    return model, best_val_acc, history

def evaluate_on_loader(model, data_loader):
    model.eval()
    y_true, y_pred = [], []
    total_loss = 0.0
    with torch.no_grad():
        for batch in data_loader:
            inputs = batch['input_ids'].to(device)
            masks = batch['attention_mask'].to(device)
            labels = batch['labels'].to(device)
            outputs = model(inputs, attention_mask=masks, labels=labels)
            total_loss += outputs.loss.item()
            preds = torch.argmax(outputs.logits, dim=1)
            y_true.extend(labels.cpu().numpy())
            y_pred.extend(preds.cpu().numpy())
    y_true = np.array(y_true)
    y_pred = np.array(y_pred)
    acc = (y_true == y_pred).mean()
    avg_loss = total_loss / len(data_loader)
    return acc, avg_loss, y_true, y_pred

# =====
# GRID SEARCH + 3-FOLD CV ON TRAIN (85%)
# =====
print(f"\n Grid Search + 3-Fold CV on TRAIN (85%)...")

LR_LIST = [1e-5, 2e-5, 3e-5]
BATCH_LIST = [8, 16]
EPOCHS_LIST = [6, 8]
MAXLEN = 128

grid_results = []
skf = StratifiedKFold(n_splits=3, shuffle=True, random_state=SEED)

for lr in LR_LIST:
    for batch_size in BATCH_LIST:
        for epochs in EPOCHS_LIST:
            print(f" → Trying: lr={lr}, bs={batch_size}, ep={epochs}")

```

```

cv_scores = []

for fold, (tr_idx, val_idx) in enumerate(skf.split(X_train_raw,
↳y_train_raw), 1):
    # Split
    X_tr_raw = X_train_raw[tr_idx]
    X_v_raw = X_train_raw[val_idx]
    y_tr_raw = y_train_raw[tr_idx]
    y_v_raw = y_train_raw[val_idx]

    # Oversample ONLY fold-train
    ros = RandomOverSampler(random_state=SEED)
    X_tr_res, y_tr_res = ros.fit_resample(X_tr_raw.reshape(-1, 1),
↳y_tr_raw)

    X_tr_res = X_tr_res.flatten() # back to list of strings

    # Tokenize (no fitting - DistilBERT tokenizer is static)
    tok_tr = tokenize_texts(X_tr_res, MAXLEN) # oversampled!
    tok_v = tokenize_texts(X_v_raw, MAXLEN) # original val

    train_ds = TextDS(tok_tr, y_tr_res)
    val_ds = TextDS(tok_v, y_v_raw)

    train_loader = DataLoader(train_ds, batch_size=batch_size,
↳shuffle=True)
    val_loader = DataLoader(val_ds, batch_size=batch_size)

    # Train (fixed epochs for fairness)
    model, val_acc, _ = train_one_run(
        train_loader, val_loader, epochs=epochs, lr=lr
    )
    cv_scores.append(val_acc)

mean_cv = np.mean(cv_scores)
std_cv = np.std(cv_scores)
print(f"    → CV Acc: {mean_cv:.4f} ± {std_cv:.4f}")

grid_results.append({
    'lr': lr, 'batch_size': batch_size, 'epochs': epochs,
    'cv_mean_acc': mean_cv, 'cv_std_acc': std_cv
})

# Select best by mean CV accuracy
grid_df = pd.DataFrame(grid_results).sort_values('cv_mean_acc', ascending=False)
best = grid_df.iloc[0]
print(f"\n Best HPs by 3-Fold CV Acc ({best['cv_mean_acc']:.4f} ±
↳{best['cv_std_acc']:.4f}):")

```

```

print({k: v for k, v in best.items() if k not in ['cv_mean_acc', 'cv_std_acc']})

# =====
# 5-FOLD CV ON TRAIN (85%) WITH BEST HPs (Final Estimate)
# =====
print(f"\n 5-Fold CV on TRAIN (85%) with best HPs...")

skf_final = StratifiedKFold(n_splits=5, shuffle=True, random_state=SEED)
cv_results = []

for fold, (tr_idx, val_idx) in enumerate(skf_final.split(X_train_raw,
↳y_train_raw), 1):
    print(f"\n--- Fold {fold}/5 ---")

    X_tr_raw = X_train_raw[tr_idx]
    X_v_raw = X_train_raw[val_idx]
    y_tr_raw = y_train_raw[tr_idx]
    y_v_raw = y_train_raw[val_idx]

    # Oversample ONLY fold-train
    ros = RandomOverSampler(random_state=SEED)
    X_tr_res, y_tr_res = ros.fit_resample(X_tr_raw.reshape(-1, 1), y_tr_raw)
    X_tr_res = X_tr_res.flatten()

    tok_tr = tokenize_texts(X_tr_res, MAXLEN)
    tok_v = tokenize_texts(X_v_raw, MAXLEN)

    train_ds = TextDS(tok_tr, y_tr_res)
    val_ds = TextDS(tok_v, y_v_raw)

    train_loader = DataLoader(train_ds, batch_size=int(best['batch_size']),
↳shuffle=True)
    val_loader = DataLoader(val_ds, batch_size=int(best['batch_size']))

    model, val_acc, _ = train_one_run(
        train_loader, val_loader,
        epochs=int(best['epochs']),
        lr=float(best['lr'])
    )
    print(f" → Fold {fold} Val Acc: {val_acc:.4f}")
    cv_results.append({'fold': fold, 'val_acc': val_acc})

# Summarize CV
cv_df = pd.DataFrame(cv_results)
mean_cv = cv_df['val_acc'].mean()
std_cv = cv_df['val_acc'].std()
print(f"\n 5-Fold CV on TRAIN (85%): {mean_cv:.4f} ± {std_cv:.4f}")

```

```

# =====
# FINAL MODEL: Retrain on FULL OVERSAMPLED TRAIN (85%)
# =====
print(f"\n Retraining final DistilBERT on FULL TRAIN (85%)...")

# Oversample FULL TRAIN only
ros_final = RandomOverSampler(random_state=SEED)
X_train_res, y_train_res = ros_final.fit_resample(X_train_raw.reshape(-1, 1),
    ↪y_train_raw)
X_train_res = X_train_res.flatten()

# Tokenize
tok_train_final = tokenize_texts(X_train_res, MAXLEN)
tok_test_final = tokenize_texts(X_test_raw, MAXLEN)

train_ds_final = TextDS(tok_train_final, y_train_res)
test_ds_final = TextDS(tok_test_final, y_test)

train_loader_final = DataLoader(train_ds_final,
    ↪batch_size=int(best['batch_size']), shuffle=True)
test_loader_final = DataLoader(test_ds_final,
    ↪batch_size=int(best['batch_size']))

# Train final model
final_model, _, final_history = train_one_run(
    train_loader_final,
    DataLoader(TextDS(tokenize_texts(X_train_raw[:500], MAXLEN), y_train_raw[:
    ↪500])),
    batch_size=int(best['batch_size']), # small internal val for
    ↪early stopping
    epochs=int(best['epochs']),
    lr=float(best['lr'])
)

# Save
SAVE_DIR = './distilbert_best_cv'
os.makedirs(SAVE_DIR, exist_ok=True)
final_model.save_pretrained(SAVE_DIR)
tokenizer.save_pretrained(SAVE_DIR)
print(f"\n Saved best model + tokenizer to: {SAVE_DIR}")

# =====
# FINAL EVALUATION ON TEST SET (15%)
# =====
test_acc, test_loss, y_true_test, y_pred_test = evaluate_on_loader(final_model,
    ↪test_loader_final)

```



```

print(f"\n FINAL TEST ACCURACY (Held-Out, Natural Distribution): {test_acc:.
↳4f}")

print("\n=== FINAL CLASSIFICATION REPORT (TEST SET) ===")
print(classification_report(y_true_test, y_pred_test,
↳target_names=class_labels, digits=4))

# Confusion Matrix
cm = confusion_matrix(y_true_test, y_pred_test)
plt.figure(figsize=(10, 8))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=class_labels, yticklabels=class_labels)
plt.title('DistilBERT (13-Class) Confusion Matrix - Final Test')
plt.xlabel('Predicted'); plt.ylabel('True')
plt.tight_layout()
plt.savefig('DistilBERT_13class_ConfusionMatrix_Test.png', dpi=300,
↳bbox_inches='tight')
plt.show()

# Learning Curves
plt.figure(figsize=(13, 5))
epochs = range(1, len(final_history['train_loss']) + 1)

plt.subplot(1, 2, 1)
plt.plot(epochs, final_history['train_loss'], 'b-o', label='Train Loss')
plt.plot(epochs, final_history['val_loss'], 'r-s', label='Val Loss')
plt.title('Loss'); plt.xlabel('Epoch'); plt.legend(); plt.grid(True)

plt.subplot(1, 2, 2)
plt.plot(epochs, final_history['train_acc'], 'b-o', label='Train Acc')
plt.plot(epochs, final_history['val_acc'], 'r-s', label='Val Acc')
plt.title('Accuracy'); plt.xlabel('Epoch'); plt.legend(); plt.grid(True)

plt.suptitle('DistilBERT Final Training')
plt.tight_layout()
plt.savefig('DistilBERT_13class_LearningCurves.png', dpi=300,
↳bbox_inches='tight')
plt.show()

```

Classes (fixed order):

- 0 → functionality issue
- 1 → user experience issue
- 2 → performance issue
- 3 → stability issue
- 4 → bug
- 5 → device issue
- 6 → customer support issue

7 → privacy issue  
8 → user interface issue  
9 → compatibility issue  
10 → installation issue  
11 → security issue  
12 → network issue

Final Splits: Train (85%): 13628, Test (15%): 2406

Grid Search + 3-Fold CV on TRAIN (85%)...

→ Trying: lr=1e-05, bs=8, ep=6

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ CV Acc: 0.8743 ± 0.0065

→ Trying: lr=1e-05, bs=8, ep=8

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',

'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ CV Acc: 0.8744 ± 0.0027

→ Trying: lr=1e-05, bs=16, ep=6

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias', 'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias', 'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias', 'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ CV Acc: 0.8707 ± 0.0033

→ Trying: lr=1e-05, bs=16, ep=8

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias', 'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias', 'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias', 'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ CV Acc: 0.8739 ± 0.0053

→ Trying: lr=2e-05, bs=8, ep=6

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ CV Acc: 0.8784 ± 0.0062

→ Trying: lr=2e-05, bs=8, ep=8

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ CV Acc: 0.8776 ± 0.0033

→ Trying: lr=2e-05, bs=16, ep=6

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ CV Acc: 0.8768 ± 0.0034

→ Trying: lr=2e-05, bs=16, ep=8

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ CV Acc: 0.8781 ± 0.0028

→ Trying: lr=3e-05, bs=8, ep=6

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

```
['classifier.bias', 'classifier.weight', 'pre_classifier.bias',  
'pre_classifier.weight']
```

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:  
['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ CV Acc: 0.8803 ± 0.0049

→ Trying: lr=3e-05, bs=8, ep=8

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:  
['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:  
['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:  
['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ CV Acc: 0.8785 ± 0.0050

→ Trying: lr=3e-05, bs=16, ep=6

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:  
['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:  
['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:  
['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']  
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ CV Acc: 0.8777 ± 0.0022  
→ Trying: lr=3e-05, bs=16, ep=8

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ CV Acc: 0.8799 ± 0.0029

Best HPs by 3-Fold CV Acc (0.8803 ± 0.0049):  
{'lr': 3e-05, 'batch\_size': 8.0, 'epochs': 6.0}

5-Fold CV on TRAIN (85%) with best HPs...

--- Fold 1/5 ---

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ Fold 1 Val Acc: 0.8866

--- Fold 2/5 ---

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ Fold 2 Val Acc: 0.9043

--- Fold 3/5 ---

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ Fold 3 Val Acc: 0.8966

--- Fold 4/5 ---

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ Fold 4 Val Acc: 0.9002

--- Fold 5/5 ---

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

→ Fold 5 Val Acc: 0.9017

5-Fold CV on TRAIN (85%): 0.8979 ± 0.0069

Retraining final DistilBERT on FULL TRAIN (85%)...

Some weights of DistilBertForSequenceClassification were not initialized from the model checkpoint at distilbert-base-uncased and are newly initialized:

['classifier.bias', 'classifier.weight', 'pre\_classifier.bias',  
'pre\_classifier.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Saved best model + tokenizer to: ./distilbert\_best\_cv

FINAL TEST ACCURACY (Held-Out, Natural Distribution): 0.9219

=== FINAL CLASSIFICATION REPORT (TEST SET) ===

|  | precision | recall | f1-score | support |
|--|-----------|--------|----------|---------|
|--|-----------|--------|----------|---------|



|                        |        |        |        |      |
|------------------------|--------|--------|--------|------|
| functionality issue    | 0.9427 | 0.9699 | 0.9561 | 797  |
| user experience issue  | 0.9281 | 0.8323 | 0.8776 | 310  |
| performance issue      | 0.9817 | 0.9273 | 0.9537 | 289  |
| stability issue        | 0.9430 | 0.9929 | 0.9673 | 283  |
| bug                    | 0.9581 | 0.9697 | 0.9639 | 165  |
| device issue           | 0.9302 | 0.8511 | 0.8889 | 141  |
| customer support issue | 0.8810 | 0.6916 | 0.7749 | 107  |
| privacy issue          | 0.9677 | 0.9278 | 0.9474 | 97   |
| user interface issue   | 0.5669 | 0.9351 | 0.7059 | 77   |
| compatibility issue    | 0.8571 | 0.8000 | 0.8276 | 45   |
| installation issue     | 0.9375 | 0.9091 | 0.9231 | 33   |
| security issue         | 0.8378 | 0.9688 | 0.8986 | 32   |
| network issue          | 0.9615 | 0.8333 | 0.8929 | 30   |
| accuracy               |        |        | 0.9219 | 2406 |
| macro avg              | 0.8995 | 0.8930 | 0.8906 | 2406 |
| weighted avg           | 0.9293 | 0.9219 | 0.9228 | 2406 |

